

# Section 7 — Evaluation Summary Explained

---

This document explains Section 7 of ResearchLens in simple Feynman-style language. It walks through every number and chart in the Evaluation Summary, using the actual implementation logic from the app.

## 1. What Section 7 is trying to do

Section 7 is a scorecard for the whole analysis pipeline. It answers questions like:

- Did we group papers into reasonable topics?
- Did we assign most papers to a topic?
- Are the discovered gaps meaningful?
- How good is the whole model overall?

The section is built from data produced by the topic, gap, and map modules. It is not a random dashboard — it computes these numbers from the actual paper assignments and gap scores.

## 2. The four core summary cards

The top cards are:

- **Dataset Size**
- **Topics**
- **Gaps**
- **High-Conf Gaps**

These are simple counts.

### Example values

- **22** papers were processed.
- **5** topics were detected.
- **1** gap was identified.
- **0** high-confidence gaps were found.

### What they mean

Think of the dataset like a room of books:

- **Dataset Size** is how many books are on the floor.
- **Topics** is how many piles the books were sorted into.
- **Gaps** is how many interesting missing connections the system found between piles.
- **High-Conf Gaps** is how many of those missing connections are very strong, score above **0.5**.

So this example says: the pipeline found 5 topic piles, only 1 gap between topics, and none of those gaps were very strong.

## 3. Topic Coherence

This is the system's measure of how consistent each topic is. In code, it is calculated like this:

```
avgCoherence = average(topic.coherence for each topic)
```

Each topic already has a **coherence** score from topic detection. The dashboard shows it as a percentage.

### Example value

- **64%** means the topics are moderately consistent.

### Feynman example

Imagine each topic is a shoe box. Topic coherence asks: are the shoes in this box really the same type?

- **100%** means every shoe in the box is clearly the same style.
- **64%** means the box is mostly consistent, but some shoes are a little different.

So with **64%**, the system is saying: "Most papers in each topic belong together, but some topics still contain slightly mixed material."

## 4. Topic Coverage

Topic Coverage measures whether the papers were actually assigned to topics. In code, it is:

```
topicCoverage = min(1, total number of paper references inside topics / total papers)
```

That means the system counts all papers that appear inside the topic groups and divides by the total paper count. It caps the result at **1.0**.

### Example value

- **100%** coverage means every uploaded paper was assigned to at least one topic.

### Feynman example

If the room has 22 books and each book is placed into some pile, then coverage is **100%**. If 2 books were left on the floor, coverage would be **91%**.

So in this output, the system did not leave any paper ungrouped.

## 5. Gap Novelty Score

Gap Novelty is the average strength of all detected gaps. In code, it is:

```
gapNovelty = min(1, average(gap.gapScore for each gap))
```

Each gap has a score between 0 and 1. A higher average means the gaps are more surprising or interesting.

### Example value

- 16% means the gaps are weak and not very novel.

### Feynman example

Think of gaps as bridges you want to build between two walls.

- A 90% gap means the wall pair is a very compelling place to build a bridge.
- A 16% gap means the gap exists, but it is not a strong or exciting opportunity.

So, with 16%, the system is saying: "I found a gap, but it is a low-novelty one, not a dramatic research opportunity."

## 6. Overall Model Quality

This is a single number that combines coherence, coverage, and gap novelty. The formula is:

```
modelQuality = round(avgCoherence * 0.4 + topicCoverage * 0.3 + gapNovelty * 0.3, 2)
```

That means:

- topic coherence counts for 40%
- topic coverage counts for 30%
- gap novelty counts for 30%

### Example value

- 60% means the model is okay. Coverage is perfect, coherence is decent, but low gap novelty keeps the score from being higher.

### Feynman example

Imagine grading a school project on three parts:

- organization = 40%
- completeness = 30%
- creativity = 30%

If organization is okay, completeness is perfect, but creativity is weak, then the combined grade is around 60%. That is exactly what this number says about the analysis.

## 7. Coherence by Topic

This chart shows each topic's individual coherence score. The app takes the topic names and the coherence values from the backend.

## Example values

- segmentation / annotated / challenge → 72%
- residual / imagenet / networks → 100%
- fine-tuned / language / unsupervised → 51%
- language / federated / dialog → 38%
- knowledge / representation / cbr-kbqa → 59%

## What it means

Each line is a separate topic. A topic at 100% is very clean and focused. A topic at 38% is messy or mixed.

## Feynman example

If each topic is a basket of apples:

- 100% means every apple in that basket tastes the same.
- 38% means the basket contains apples and oranges.

So the system is telling you which topics are tight clusters and which ones may need better separation.

## 8. Gap Score Distribution

The section also shows how the gap scores are spread across buckets:

- 0.0-0.2
- 0.2-0.4
- 0.4-0.6
- 0.6-0.8
- 0.8-1.0

Each bucket counts how many gaps fall into that range.

## Example distribution

- 1 gap in 0.0-0.2
- 0 gaps in every higher bucket

That means the only gap found is very weak.

## Feynman example

Imagine measuring how big each crack is in a wall.

- Most cracks are tiny if they are in 0.0-0.2.
- A strong, actionable crack would be in 0.6-1.0.

The app says: if most gaps were in 0.6-1.0, these would be strong research opportunities. But this example has the gap in the weakest bucket.

## 9. What you should take away

In plain language:

- the system processed all 22 papers, so the dataset is complete.
- it found 5 topic groups, which is a reasonable number.
- topic coverage is perfect, so no paper was left out.
- topic coherence is moderate, so some topics are good while others need improvement.
- gap novelty is low, meaning the discovered gap is not a very exciting research opportunity.
- model quality is okay, driven down by the weak novelty even though coverage is perfect.

## 10. Why this is useful

This section helps you decide whether the analysis is worth trusting.

- if coherence is low, the topics may need better clustering,
- if coverage is low, some papers are being ignored,
- if gap novelty is low, the system is not finding new ideas,
- if model quality is low, the overall result is still weak.

So Section 7 is not just decoration. It is a diagnostic panel that tells you whether the pipeline worked well or whether the analysis needs improvement.

---

### File location

Saved as `research-evaluation-explanation.md` in the project root.